STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-GOOD-ON-OTHER-SIDE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4976-4986 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4988-5003, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 35 of queue_sdga.json (pb=770, stmt 4976-4986, proof 4988-5003).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$, $\mathcal{B}$ be differential graded $\mathcal{O}$-algebras. Let $\mathcal{N}$ be a differential graded $(\mathcal{A}, \mathcal{B})$-bimodule. If $\mathcal{N}$ is good as a left differential graded $\mathcal{A}$-module, then we have $\mathcal{M} \otimes^{\mathbf{L}}_{\mathcal{A}} \mathcal{N} = \mathcal{M} \otimes_{\mathcal{A}} \mathcal{N}$ for all differential graded $\mathcal{A}$-modules $\mathcal{M}$.*

Proof. Let $\mathcal{P} \to \mathcal{M}$ be a quasi-isomorphism where $\mathcal{P}$ is a good (right) differential graded $\mathcal{A}$-module. To prove the lemma we have to show that $\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \to \mathcal{M} \otimes_{\mathcal{A}} \mathcal{N}$ is a quasi-isomorphism. The cone C on the map $\mathcal{P} \to \mathcal{M}$ is an acyclic right differential graded $\mathcal{A}$-module. Hence $C \otimes_{\mathcal{A}} \mathcal{N}$ is acyclic as $\mathcal{N}$ is assumed good as a left differential graded $\mathcal{A}$-module. Since $C \otimes_{\mathcal{A}} \mathcal{N}$ is the cone on the maps $\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \to \mathcal{M} \otimes_{\mathcal{A}} \mathcal{N}$ as a complex of $\mathcal{O}$-modules we conclude. □

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